Band-1 Hadamard Order 8644 – Miyamoto (Kronecker Seed H_2160=4x540 + C_2162); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_540 Paley-II PASS; H_2160 Kronecker PASS; C_2162 PASS; H_8644 Miyamoto PASS; classical SPK FAIL -> Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-16

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8564_PaleyI`; `2026-09-16_Band1_Hadamard_8516_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8644_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2160 and 8644 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed $H_{2160} = \text{Kronecker}(H_4, H_{540})$, $H_{540} = \text{Paley-II}(q = 269)$, and symmetric Paley conference C_{2162} over GF(2161) ($q = 2161 \equiv 1 \pmod 4$). Classical SPK for 8644: Paley-I **FAIL**s ($q = 8643 = 3{\cdot}43{\cdot}67$ NOT prime); Paley-II **FAIL**s ($q = 4321 = 29 \cdot 149$ NOT prime power – no matrix attempt); Kronecker **FAIL**s (4×2161). Therefore order **8644 is a Miyamoto-only bonus NOT in SPK**. Seed **2160** is **IN SPK** via Kronecker of classical seeds. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8564 IN SPK via Paley-I** (seeds **2140+8564**). Prior SPK seeds **2140+8564 / 1064+2128 / 2112 / 2088+8356 / 2028+8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **8644/8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2160** folded into ONBOX/SPK when Kronecker verified; **8644** never folded (Miyamoto alone never folds SPK).

Abstract

Finite constructions of orders **2160** and **8644**.

- H_{540} = Paley-II ($q = 269$) – Kronecker factor (loaded prior / verified)
- H_{2160} = Kronecker(H_4, H_{540}) – preferred Miyamoto seed (Paley-II $q = 1079$ FAIL; Paley-I $q = 2159$ FAIL; Double FAIL)
- C_{2162} = Paley conference over GF(2161), $q = 2161$
- H_{8644} = Miyamoto(H_{2160}, C_{2162})
- Paley-I for 8644: **SKIP** ($q = 8643 = 3 \cdot 43 \cdot 67$ NOT prime)
- Paley-II for 8644: **SKIP** ($q = 4321 = 29 \cdot 149$ NOT prime power; no matrix attempt)
- Kronecker for 8644: **SKIP** ($4 \cdot 2161$ not admissible)

SPK: 8644 Miyamoto-only NOT in SPK. Seed 2160 enters SPK via Kronecker. Prior SPK **2140+8564 / 1064+2128 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8644.py`

Results. overall_pass=true; units 164/164; wall 531.5327s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8644 = 4 \cdot 2161$; 2161 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2160} and C_{2162} (Paley conference over prime field $q = 2161$).
- Seed: Kronecker 4×540 with H_{540} = Paley-II($q = 269$) (preferred); Paley-II $q = 1079$ FAIL; Paley-I $q = 2159$ FAIL; Double FAIL.
- SPK: Paley-I $q = 8643$ **FAIL**; Paley-II $q = 4321$ **FAIL** (NOT prime power); Kronecker **FAIL** -> **Miyamoto-only NOT in SPK**.
- Prior: 8564 IN SPK via Paley-I shipped; this ship $q = 2161 \rightarrow n = 8644$ Miyamoto-only.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_4 Sylvester	Sylvester	0	0.0001
H_540 Paley-II	Paley-II q=269 (loaded/verified)	0	0.0869
H_2160 Kronecker	Kronecker(H_4, H_540)	0	4.4227
C_2162	Paley conf GF(2161) q=2161	0	6.5133
H_8644 Miyamoto	Miyamoto(H_2160 Kronecker, C_2162)	0	516.9142
H_8644 Paley-I	SKIP q=8643=3*43*67	–	–
H_8644 Paley-II	SKIP q=4321=29*149 (no matrix)	–	–
H_8644 Kronecker	SKIP 4*2161 not admissible	–	–

Object	Method	max|HH^T-nI|	Wall (s)

SPK closure note

Seeds include prior SPK ships through **8564** lineage (**2140+8564**; **1064+2128**; **2112**; **2088+8356**; **2028+8116**; **1996**; **1932**; **7604**; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 8644/8516/8452/7988/7732/7492/7156/71
+ verified **2160** (Kronecker 4x540); **exclude** Miyamoto-only bonuses including **8644**. Next constructive gap **2052** (OPEN).

Quantity	Value
8644 in SPK	**False** (Miyamoto-only)
2160 in SPK	**True** (Kronecker 4x540)
540 in SPK	**True** (Paley-II q=269)
2140 / 8564 / 1064 / 2128 in SPK	**True** (prior)
8644 / 8516 / 8452 / 7988 / 7732 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture * no Clay-closed * no finished-ToE * no RH / P!=NP / YM / Frankl / NS-solved / AC-solved * gap 2052 remains OPEN * finite constructions of 2160+8644 ONLY * Miyamoto alone never folds into SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_8644_miyamoto/`
Artifacts: `H540_pm1.npz`, `H2160_kronecker_pm1.npz`, `C2162_pm1.npz`, `H8644_miyamoto_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

Next optional

q=2197->n=8788 PaleyII(q=1097) OR q=2209->n=8836 PaleyI(q=2207)+Double(PaleyI half=1104) OR q=2221->n=8884 PaleyII(q=1109) OR method-attack SPK gap 2052 OR Track B / Object 1 meter await